Project 3A – Dictionaries and Sets

Instructions: Using Python, create a file called" Project3A-Dictionaries_Sets" and follow the directions below. Take pictures of your executed program and submit all files – Python files, dictionaries, sets, and pictures of your executed programs.

📘 Starter Data (Students must create these)

Dictionaries

Dictionary 1 — NFL Teams (Includes Philadelphia Eagles)

python

```
nfl_teams = {
    "Eagles": "Philadelphia",
    "Bills": "Buffalo",
    "Dolphins": "Miami",
    "Patriots": "New England",
    "Jets": "New York"
}
```

Dictionary 2 — NBA Teams (Includes Philadelphia 76ers)

python

```
nba_teams = {
    "76ers": "Philadelphia",
    "Lakers": "Los Angeles",
    "Celtics": "Boston",
    "Bulls": "Chicago",
    "Heat": "Miami"
}
```

Sets

Set 1 — Pennsylvania Cities (Subset)

```
python
pa_cities = {"Philadelphia", "Pittsburgh", "Harrisburg", "Erie"}
```

Set 2 — Mid-Atlantic Cities (Superset)

```
python
midatlantic_cities = {
 "Philadelphia", "Pittsburgh", "Harrisburg", "Erie",
 "Baltimore", "Washington", "Newark"
}
```

Note: pa_cities is a **subset** of midatlantic_cities. midatlantic_cities is a **superset** of pa_cities.

🧠 **Student Tasks**

1. Create the dictionaries and sets

2. Add items to each dictionary and set

```
nfl_teams["Commanders"] = "Washington"
nba_teams["Warriors"] = "Golden State"
pa_cities.add("Allentown")
midatlantic_cities.add("York")
```

3. Find the union of the sets

4. Find the intersection of the sets

5. Loop through each dictionary and print items

6. Print only the keys of the NFL and NBA

7. Print only the values of the NFL and NBA

8. Merge and print the dictionaries

9. Update one dictionary with the other and print the results

10. Find the symmetric difference of the sets

11. Check and print the subset and superset

📊 **Rubric (100 points)**

Category	Points	Description
Dictionary creation & modification	20	Correct creation, adding, deleting, printing keys/values
Set creation & operations	20	Correct union, intersection, symmetric difference
Loops	10	Correct iteration through dictionaries
Merge & update operations	20	Correct use of ` and =`
Subset & superset checks	20	Correct use of <= and >=
Code readability & comments	10	Clear formatting and comments